Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science **West Lafayette, IN**
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics; GPA 3.4/4 **May 2028**

PROFESSIONAL EXPERIENCE

Boston Dynamics **Waltham, MA**
Sensors and Perception Co-Op – Atlas | *Python, MuJoCo, Flask, SQLite, CATIA, Linux* **July 2026 – Present**

- Estimated impact energy from Khatib's operational-space effective mass and an OLS velocity fit, fed by a forward-kinematics pipeline propagating 6-DOF floating-base poses through quaternions and Rodrigues' rotations.
- Developed a physics-based MuJoCo impact simulation and a custom test around a pendulum-mounted six-joint arm that rotates about all three axes, hand-deriving its Jacobians to validate the effective-mass model.
- Built an impact forensics pipeline and Flask/Plotly dashboard that ETLs ~900 robot test logs into SQLite, giving engineers a 3D heatmap of ~28k impacts on the robot's mesh and per-event waveforms during live testing.
- Modeled hardware thermal behavior in Python, fitting heating curves with Savitzky-Golay filtering and multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.
- Designed fixtures, mounts and shear features in CATIA, optimizing their geometry through pin-shear failure analysis.

Caterpillar (via Purdue Data Mine) **West Lafayette, IN**
Machine Learning Researcher | *Python, Snowflake, Power BI* **January 2026 – May 2026**

- Cut material usage 4.7% by forecasting next-day primer, topcoat and solvent consumption with LSTM and XGBoost models trained on five years of paint-line data, with Monte Carlo confidence intervals on every forecast.
- Consolidated five years of noisy legacy files into one daily series, contributing to a 3.3% cut in cycle time.

Chortos Lab – Soft Adaptive Systems and Structures **West Lafayette, IN**
Undergraduate Researcher | *Python, LabVIEW, MATLAB, Fusion 360* **August 2025 – May 2026**

- Automated LabVIEW and MATLAB acquisition from a Keyence laser displacement head via an NI USB-6001, and wrote a Python computer-vision tracker measuring cantilever deflection to 10 µm against an in-frame micrometer scale.
- Formulated, synthesized and direct-ink-write printed ionic dielectric gels, tuning their chemistry against force-voltage tests on 15 custom Fusion 360 fixtures to raise electromechanical force output by over 50%.

Emerson – ASCO **Florham Park, NJ**
Mechanical and Manufacturing Engineering Intern | *Creo, CNC, GD&T* **May 2025 – August 2025**

- Designed 30+ rotary valve components in Creo, verified by hand stress analysis, and machined pistons to ±0.05 mm.
- Root-caused actuator deadband from cycle-test data and wrote an assembly procedure that cut it by over 10%.

PROJECTS

Purdue ASME – Biomechanics **West Lafayette, IN**
Active Orthosis – Electronics Team Lead **April 2024 – May 2026**

- Designed a 1 kHz feed-forward PI velocity controller with anti-windup in C++17 on an ESP32-S3, tuned from a first-order motor model to hold simulated speed error to 0.8%, guarded by 10 latching fault monitors and a watchdog.
- Cut EMG-to-motion latency from 103 ms to 43 ms with a Bayesian EMG estimator, validated in closed-loop simulation through the controller and a simulated arm on 24 real forearm contractions.
- Led a 5-engineer team redesigning the 4-channel EMG board in KiCad with an INA333 instrumentation front end, right-leg drive and ISO7763 motor isolation, raising day-to-day gesture decoding from 31% to 87% in emulated tests.

Asteroid Hopping Robot – Independent Research
Purdue University & Universidad Carlos III de Madrid **October 2026 – Present**

- Building a three-legged asteroid hopper that reshapes its legs between hops, designing push-off control optimized for a target launch speed, knee impedance control to absorb landings and reaction-wheel control to reorient mid-flight.
- Sizing each hop in MuJoCo by comparing risk-limited policies that trade distance against the chance of escaping the asteroid or tipping over, and developing a pendulum rig to test landings.

Legged Robot State Estimation and Control
Personal Project **August 2026 – October 2026**

- Built a contact-aided error-state Kalman filter in C++/Eigen that fuses a simulated quadruped's 1 kHz IMU with leg kinematics, estimating base velocity to 0.013 m/s RMSE against 1.38 m/s for IMU-only.
- Verified every Jacobian against finite differences, then ran a Raibert trot controller and a QP balance controller on the estimate instead of ground truth, walking 5 seeds with zero falls and velocity tracking within 0.004 m/s.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, LabVIEW, Eigen, MuJoCo, FreeRTOS, GoogleTest, KiCad, CATIA, Creo, Linux
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Biliteracy)
- **Clubs & Hobbies:** Club Swimming, House Band, Lifting, Hiking, Guitar, Saxophone